

Using Learning from Demonstration (LfD) to Train Robots for Complex Poultry Processing Tasks

BY MICHAEL PARK

An increasing number of robots are being deployed across manufacturing, agriculture, and healthcare to address labor shortages or to improve the efficiency of manual operations. These robots are usually made with all-purpose generic hardware and are programmed for multiple purposes, which often include the ability to self-navigate in unstructured environments or make complex decisions.

Currently, programming a robot is difficult and time-consuming. It is at least a two-step process where an engineer first needs to understand a task and then translate it into a code that a robot can understand. For instance, for a breast deboning task in poultry processing, an engineer must learn how an expert deboner inserts and guides his or her knife to make a cut and translate that process into a code, which often is unnatural and nonintuitive. In addition, every movement of a robot must be custom-tailored for accomplishing different tasks, which requires an expert engineer to program the robot.

A lot of research has been conducted since the 1980s to minimize, or even eliminate, this difficult step by letting users train the robot to fit their needs by utilizing expert knowledge from the user, in the form of demonstrations. This is a paradigm called robot learning from demonstration (LfD) or robot programming by demonstration. LfD provides a very efficient programming interface where even non-expert users can easily and quickly teach a robot how to do various tasks through either visual demonstrations or physical guidance.

LfD is not a simple record and play technique, and its power goes beyond providing an easy interface. In fact, LfD allows users to encode human intuition. By observing several demonstrations, a robot can learn how to

generalize a skill and flexibly apply it to varying environments.

Modern poultry processing plants have increasingly been automated. However, there remains several tasks that require human experts such as hanging, cone loading, or trimming. Even for the automated tasks, the machines need to be manually adjusted to accommodate the size variance in birds to minimize yield loss. By observing multiple demonstrations of skilled workers performing such tasks, a robot can obtain

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Researchers in the Georgia Tech Research Institute's (GTRI) Agricultural Technology Research Program have applied LfD to teach a robot to debone chicken breasts. A custom knife tool that includes ArUco tags and an

inertial measurement unit was created to collect position, velocity, and acceleration data from experts' demonstrations as shown in Figure 1. The images of the birds were also collected, and machine learning was used to learn the association between the bird's external features and the expert's knife trajectory. The learned deboning strategy was deployed on a 6 degrees of freedom robot manipulator, and the manipulator was able to perform a cut with a much more natural and smoother

motion compared to the motion programmed by engineers. The robot was also able to cut birds with varying sizes without any adjustments as long as similar looking and similar sized birds were captured in the collected data. Figure 2 shows the resulting cut.

The next goal is to pair the LfD technique with a real-time force feedback control to create a robust robotics deboning system that not only can autonomously adapt to varying sizes and shapes of the birds, but also make an optimal cut by continuously adjusting the knife path to find the joints.

LfD provides an easy and intuitive way to program robot behaviors, potentially saving development time and costs. It can also add flexibility to robots by encoding human intuition and knowledge for a more robust operation. ♥

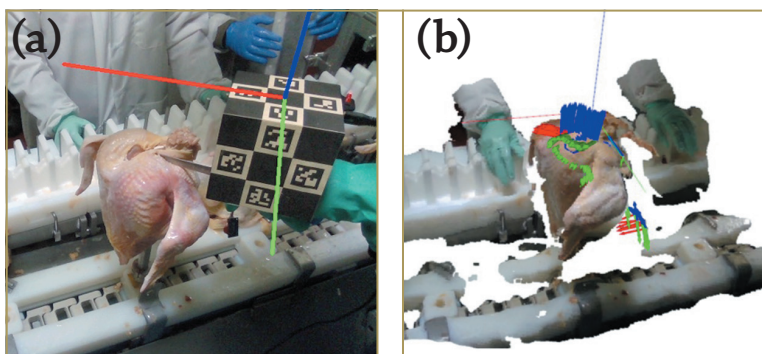


Figure 1. (a) A custom knife tool that includes ArUco tags and an inertial measurement unit was created to collect position, velocity, and acceleration data from experts' demonstrations. (b) Reconstructed point cloud showing knife pose throughout cutting trajectory.



Figure 2. The learned deboning strategy was deployed on a 6 degrees of freedom robot manipulator, and the manipulator was able to perform a cut with a much more natural and smoother motion compared to the motion programmed by engineers.

human-level flexibility in handling variations and uncertainties in bird sizes, shapes, and weights. In addition, the robots can easily be repurposed to carry out different tasks without significant overhead.